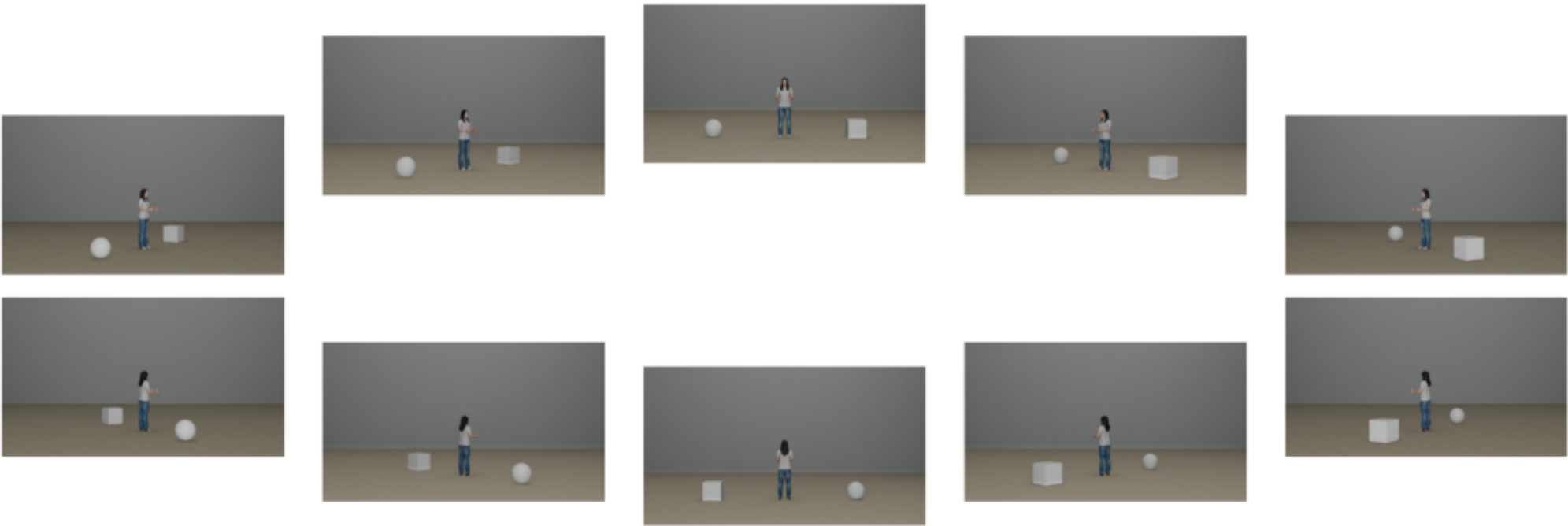


(a)



(b)

Embodiment Tokens



1. Extract keypoints

$\Delta x = 192 - 172 = 20$
 $\Delta y = 81 - 53 = 28$
 $\theta = \arctan2(-(28), 20) \times \frac{180}{\pi} + 360$
 $= 305.55^\circ$

2. Compute orientation

<POSE_START> <SHOULDER_L>
<X_172> <Y_53> <SHOULDER_R> ...
<ORIENT_START> <YAW_6> <TORSO_W_2>
<ORIENT_END> <POSE_END>

3. Generate tokens

Rotation Tokens



1. Extract (X, Y) coordinates

 **OrientAnything**
azimuth: 222
(180 - 222)%360 = 318 ≈ 315°
azimuth: 267
(180 - 267)%360 = 273 ≈ 270°

2. Extract orientation

<OBJECT_START> <OBJ_PERSON>
<X_-4> <Y_6>
<AZIMUTH_315> <OBJECT_END>
<OBJECT_START> <OBJ_FURNITURE>
<X_-98> <Y_38>
<AZIMUTH_270> <OBJECT_END>

3. Generate tokens

(c)

Atomic

Q:
Estimate this person's pose.

A:
<POSE_START> <SHOULDER_L> <X_172> <Y_53>
<SHOULDER_R> ... <ORIENT_START> <YAW_6>
<TORSO_W_2> <ORIENT_END> <POSE_END>

Chain of Thought

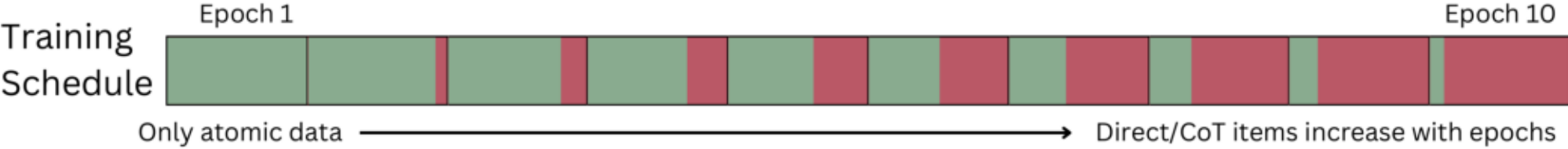
Q:
A person and a chair are shown in the image. Which side of the person is the chair on from the person's perspective? To answer this question, let's think through it step by step...

A:
Left

Direct

Q:
A person and a chair are shown in the image. Which side of the person is the chair on from the person's perspective?

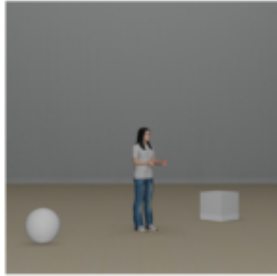
A:
Left



(d)

Synthetic

Naturalistic



Which side of the person is the cube on from the person's perspective?



Isle Dataset

Goral et al. 2025



From the perspective of the humanoid minifigure, where is the object located relative to it?



COCO

Lin et al. 2015



Which side of the person is the bike on from the person's perspective?

3DSRBench

Ma et al. 2025



Orientation

Q: Is the stop sign on the left or right side of the man on the bicycle?
A: On the right side.